

Cyclistic- Case Study Report

Ivan Bjelica

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For this case study, I analyzed over a year of Cyclistic bike-share trip data (Chicago's Divvy system) to understand how annual members and casual riders use the service differently, with the goal of informing a marketing strategy to convert casual riders into members. Using Python and pandas, I cleaned and validated 8.9M+ trip records, resolved data quality issues including station ID inconsistencies across years and DST-related timestamp anomalies, and built a star-schema dataset for analysis in Tableau.

The resulting dashboard revealed clear behavioral differences between the two groups: casual riders take longer, more variable trips concentrated on weekends and summer months near lakefront destinations, while members ride steadily year-round with sharp weekday commute peaks.

These patterns pointed to three actionable recommendations - geographically and seasonally targeted campaigns, weekend-timed messaging, and value framing around leisure flexibility rather than commuting - while also surfacing a clear boundary in what trip data alone can answer, since understanding why casual riders would convert requires additional data like surveys or pricing-response tracking.

ASK

Business Task

Cyclistic's director of marketing, Lily Moreno, believes the company's future growth depends on converting casual riders into annual members, since Cyclistic's finance team has determined that annual members are significantly more profitable than casual riders. To design an effective conversion campaign, the marketing team first needs to understand the behavioral differences between these two rider segments. This analysis addresses that need directly: how do annual members and casual riders use Cyclistic bikes differently? The findings will serve as the evidentiary foundation for a data-driven marketing strategy aimed at converting casual riders into annual members, which the executive team must approve before implementation.

PREPARE

Data location and organization

The dataset consists of individual monthly trip files (CSV format) covering January 2025 through July 2026, downloaded directly from Divvy's public data repository. Files were organized into a structured local project directory by year, with raw files kept separate from working copies to preserve an unmodified original of the source data throughout the cleaning process.

Data credibility (ROCCC)

The data has been made available by Motivate International Inc. under this [license](#).

The dataset is Reliable and Original, as it is collected directly by Divvy's own bike-share system rather than sourced from a third party. It is reasonably Comprehensive, capturing every ride at the individual trip level across the full station network. It is largely Current, covering all of 2025 and January-July 2026; however, the 2026 portion is year-to-date only, which limits direct month-over-month comparisons for the second half of the year and is accounted for explicitly in the Process and Analyze stages. The data is publicly Cited and governed by a documented license agreement from Motivate International Inc.

Verifying data integrity

Before cleaning began, the 2025 and 2026 datasets were compared for structural consistency - confirming matching column names, order, and data types across both years - to ensure they could be safely combined without silent schema mismatches. File counts were checked against expected monthly coverage, and a sample of records was spot-checked against known ride patterns to confirm the data loaded correctly.

The data contain individual ride-level records, including:

- ride_id – unique identifier for each ride
- rideable_type – type of bicycle used
- started_at – ride start timestamp
- ended_at – ride end timestamp
- start_station_name / start_station_id
- end_station_name / end_station_id
- start_lat / start_lng
- end_lat / end_lng
- member_casual – whether the rider was a member or casual rider

Relevance to the business task

The `member_casual` field directly enables comparison between the two rider segments central to this analysis. The `started_at/ended_at` timestamps support the *when* and *how long* dimensions of the analysis, while the station name, ID, and coordinate fields support the *where* dimension - together, these fields provide the behavioral detail needed to answer how members and casual riders use Cyclistic differently.

PROCESS

Python and pandas were used for the data preparation stage because they allow large numbers of records to be inspected and transformed consistently.

The 2025 and 2026 datasets were checked to ensure that they contained the same columns, data types. This helped confirm they can be combined without unexpected structural differences

The data were cleaned and validated to the extent possible given the limitations of the source data.

The notebook contains the sequence of:

1. Loading the monthly CSV files
2. Combining the yearly datasets
3. Checking column consistency
4. Inspecting data types
5. Checking missing values
6. Calculating ride duration
7. Investigating negative durations
8. Investigating daylight-saving-time effects
9. Checking duplicate records
10. Investigating duplicate ride IDs
11. Removing confirmed duplicate rides
12. Creating the final analysis dataset
13. Creating additional tables for Tableau analysis

Data-quality issue	Investigation / action
Different monthly files	Loaded and combined using Python
Column consistency	Compared columns between yearly datasets
Data types	Inspected and converted timestamps to datetime
Missing values	Counted and calculated percentages for each column
Duplicate rows	Checked at yearly and combined-dataset level
Duplicate ride IDs	Investigated individually
35 duplicate ride IDs	Confirmed as the same rides appearing across monthly boundaries; removed
Ride duration	Calculated from start/end timestamps
Negative durations	Investigated rather than automatically deleted
Fall DST transition	29 affected records identified and corrected
Spring DST transition	Identified and retained because exact elapsed time could not be determined confidently
Very long rides	Investigated as potential outliers; not automatically removed
Missing station data	Retained because other ride-level information remained useful
New year riders data (2024)	Excluded 59 rows with started_at prior to 2025-01-01 (rides spanning the 2024/2025 year boundary)

The final dataset was checked for:

- consistent column structure
- appropriate data types
- valid datetime fields
- missing values
- duplicate ride IDs
- negative ride durations
- unusually short rides
- unusually long rides
- reasonable distributions of ride duration
- valid rider categories
- valid bicycle categories

Ride duration was also checked using descriptive statistics and specific thresholds. The cleaned dataset was then exported for use in Tableau.

ANALYZE

Analysis Summary

How do annual members and casual riders use Cyclistic bikes differently?

The analysis reveals clear differences in how annual members and casual riders use Cyclistic bikes across **time, duration and location**.

When do they ride?

Both groups show strong seasonality, with ride volumes increasing during the warmer months and declining during winter. However, the pattern differs substantially between user types.

Members show more consistent weekday activity throughout the year, while casual riders are more concentrated on weekends and during the summer months. Seasonal composition reinforces this difference, with members accounting for approximately 80% of rides during winter while the proportion of casual rides increases considerably during summer.

Hourly activity also reveals distinct patterns. Members show pronounced peaks around **8 AM and 5 PM**, while casual ridership increases throughout the daytime and reaches its highest level around the afternoon. When weekday and weekend activity are examined separately, the pronounced hourly peaks among members are largely associated with weekday activity. Casual riders, particularly on weekends, show a more evenly distributed pattern throughout the day.

How long do they ride?

Casual riders generally have longer ride durations than members. However, the difference between the two groups depends strongly on how duration is measured. Casual riders have a substantially higher mean duration and considerably greater variability, while the difference in median duration is much smaller.

This indicates that a relatively small number of very long rides contribute significantly to the difference in average duration. Casual ride durations also vary considerably more than member durations, suggesting a wider range of riding behaviors. The available data cannot determine the specific purpose of these trips, but the greater variability is consistent with casual riders having a less uniform usage pattern.

Where do they ride?

The geographic analysis provides another clear distinction between the two groups. **Casual rides are more concentrated around lakefront locations**, including areas around Navy Pier and Lake Shore Drive, while member activity is more concentrated in the central/downtown area.

These spatial differences are consistent with different patterns of usage. Casual riders appear more concentrated around lakefront destinations, while members show greater activity within the central part of the city's cycling network.

Overall finding

Taken together, the analysis suggests that **members and casual riders use Cyclistic for different patterns of activity**. Members demonstrate a more consistent, weekday-oriented usage pattern with pronounced morning and afternoon peaks and shorter, less variable rides. Casual riders show stronger seasonality, greater weekend and daytime activity, longer typical rides, and greater variation in ride duration. Their rides are also more concentrated around lakefront areas.

The data therefore suggest that casual riders represent a distinct user segment rather than simply less-frequent members. Understanding when, where and how they ride provides a basis for developing membership strategies tailored to their usage patterns.

SHARE

[Tableau Dashboard link](#)

Key Findings:

WHEN: Casual riders are more seasonal and weekend-oriented.

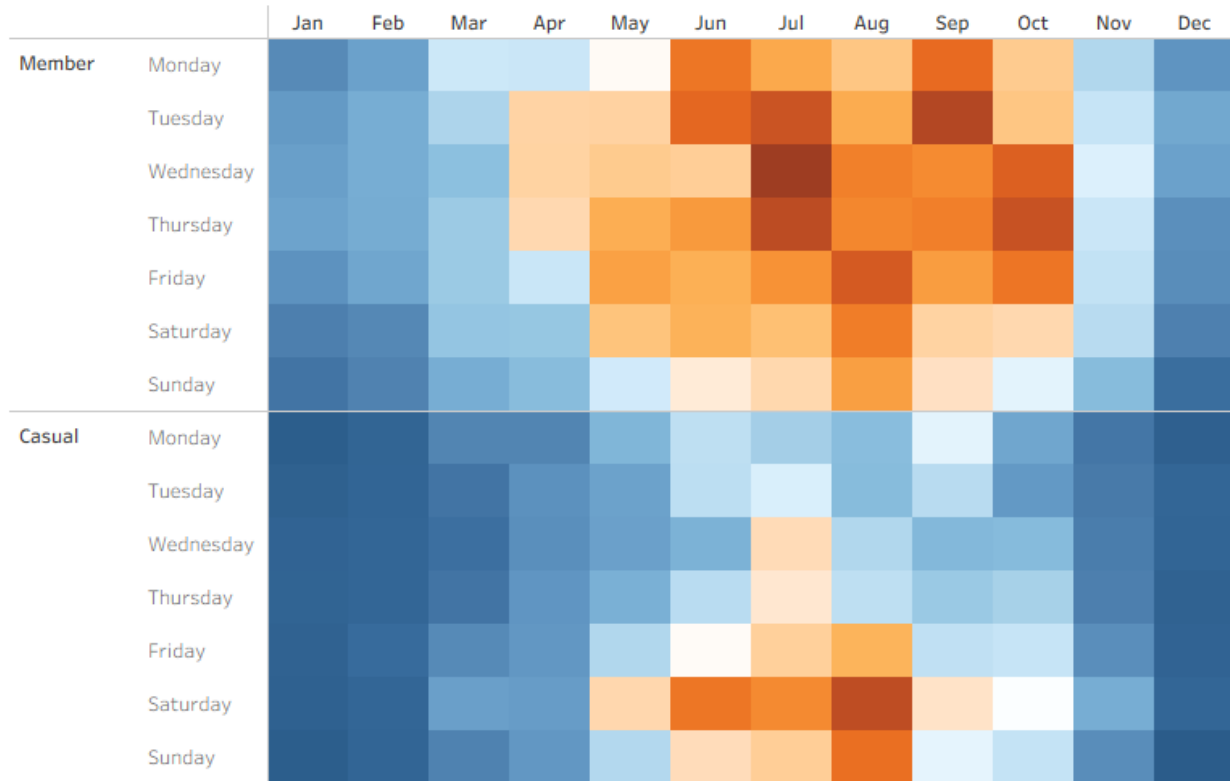
HOW LONG: Casual rides are longer and substantially more variable.

WHERE: Casual rides are concentrated around lakefront locations.

The visualizations below prioritize clarity and direct comparison over technical detail. Each chart places member and casual riders side by side, allowing the behavioral differences to be seen at a glance rather than requiring the reader to interpret raw statistics.

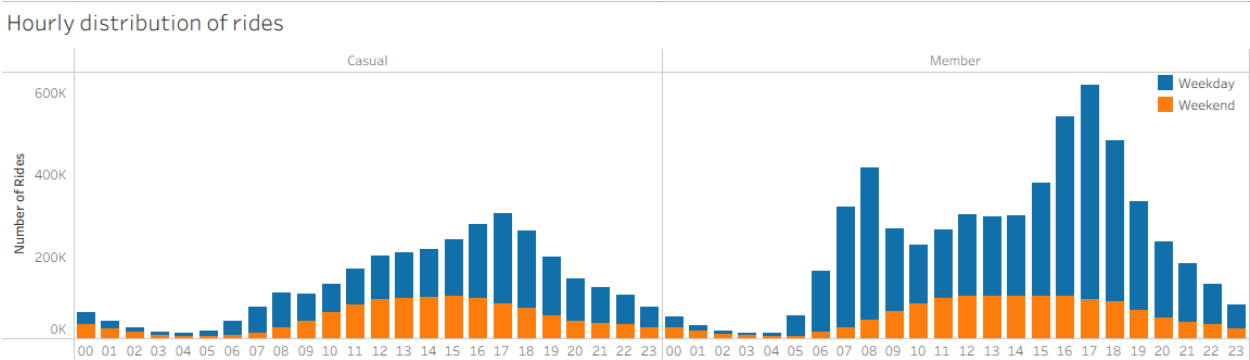
Visual 1 — When do the bikes come out?

When do the bikes come out?



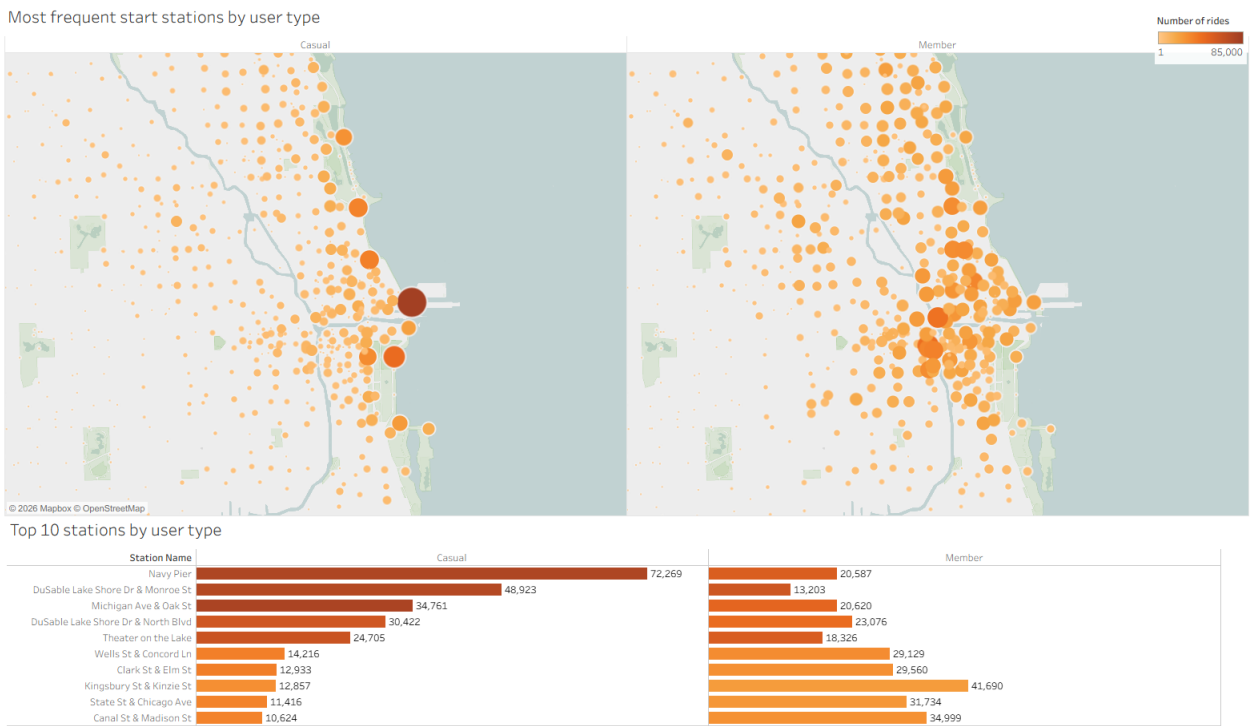
This heatmap shows ride volume by day of week and month, split by rider type. Member activity (top) is relatively steady with a modest summer increase, while casual activity (bottom) is heavily concentrated in the summer months, particularly on weekends - visually confirming the seasonality and weekend-orientation described in the Analyze section.

Visual 2 — Hourly distribution of rides



This chart breaks down ride volume by hour of day, separated into weekday and weekend activity for each rider type. Members show a clear two-peak commute pattern on weekdays (around 8 AM and 5 PM); casual riders show a broader, more gradual rise through the afternoon with a comparatively larger share of weekend activity, reinforcing that casual usage follows a leisure pattern rather than a commute pattern.

Visual 3 — Most frequent start stations by user type



This map and chart show the ten most frequent start stations for each rider type. Casual rides concentrate heavily around lakefront and tourist-oriented locations - Navy Pier alone accounts for over 72,000 casual rides, more than triple its member volume - while member activity is more evenly

distributed across downtown and residential stations. This spatial contrast supports the recommendation to geographically target lakefront stations in conversion campaigns.

ACT

Top 3 recommendations

1. Target digital marketing geographically and seasonally, concentrated at lakefront stations during peak casual season (late spring through summer).

Your geographic and seasonal findings show casual activity concentrates around Navy Pier/Lake Shore Drive and spikes in summer - this is a directly actionable targeting parameter (e.g., geo-fenced social/digital ads around those specific stations, timed to when casual ridership is highest) without requiring any new data.

2. Time membership-conversion messaging around weekend and midday usage windows, since that's when casual riders are actually on the app/bikes, rather than mimicking commute-hour targeting that works for retaining members. In-app or station-based prompts (e.g., a QR code or notification at dock/undock) timed to weekend afternoons would reach casual riders in the moment they're actively using the service.

3. Frame membership value around flexibility/leisure use rather than commuting, since casual riders show longer, more variable ride durations - the current single-ride/day-pass pricing may already suit occasional leisure trips, so the campaign should test messaging around cost savings for *frequent* leisure use (e.g., "if you ride more than X times a summer, membership pays for itself") rather than the commuter-savings angle likely used to attract members.

Additional data/analysis needed to directly answer "why would they convert"

- **Rider survey or exit-intent data** - directly asking casual riders (e.g., at point of purchase, or via app prompt) what would make them consider a membership. This is the single most direct way to answer question 2, and nothing in trip data alone substitutes for it.
- **Pricing/elasticity data** - how casual riders' usage frequency correlates with what they'd pay for membership; A/B testing different membership price points or trial offers.
- **Repeat-rider tracking** (anonymized, respecting the privacy limitation already noted in Prepare) - identifying whether the same casual riders return repeatedly (frequent casual users are a stronger conversion target than one-time tourists) vs. one-off users. Currently we can't distinguish a tourist who rode once from a regular commuter who

just hasn't converted yet, and those two groups need completely different marketing approaches.

- **Campaign response data, once a pilot runs** - after testing recommendation 1-3 above, tracking actual conversion rate by channel/message would close the loop and let a future analysis directly measure what drove conversion, turning this from a hypothesis-generating study into a causally-informed one.
- **Competitor/market context** - whether casual riders are using other transit or bike-share options, which would clarify whether "convert to member" is competing against walking/transit/rideshare or against a competing bike-share service.